



Baseline

First record of pyroplastic and partially burnt plastic litter along South African shores

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ARTICLE INFO

Keywords:

Plastic pollution
Coastal plastic debris
Environmental monitoring
Intertidal habitats
Marine litter

ABSTRACT

As plastic pollution has accumulated in the natural environment over the last half-century, so too has the body of literature on plastic contamination assessment. Despite this growing research, new, and cryptic forms of plastic debris continue to emerge, with limited data on their composition and abundance. Among these novel pollutants is a type of marine litter derived from the burning or melting of manufactured plastics. In this study, we report for the first time the presence of burned plastic forms at selected intertidal sandy habitats along the South African coast. We examined both pyroplastics—molten items with rock-like properties due to environmental weathering—and partially burned plastics, which retain features of the original manufactured objects. Surveys conducted across 22 sites spanning over 2000 km revealed this new type of litter at 19 locations, predominantly composed of polyethylene (37 items), polyethylene terephthalate (12 items), polypropylene (5 items) and polystyrene (3 items), with an average weight of 5.4 g per item. These findings highlight the widespread presence of burned plastics in coastal environments in South Africa and emphasize the need for further research into their ecological impacts.

Plastics are an integral part of modern society, with their applications constantly expanding (Europe, 2022). As plastic production continues to soar, excessive use and poor management at both pre- and post-consumer stages have led to global contamination; plastic litter is now found in a wide variety of ecosystems, from urban areas to remote locations (Rangel-Buitrago and Neal, 2023; Velez et al., 2019; Windsor et al., 2019). Most of the plastic litter originates from land, entering waterways, and wastewater systems (e.g., Harris et al., 2021). Once in the marine environment, plastic debris can disperse via wind and currents, accumulating in coastal areas and beyond (Tong et al., 2021; Velez et al., 2020). Evidence highlights the numerous harmful effects of plastic contamination—defined as presence of a substance where it should not be or at concentrations above background levels—on a variety of species in marine, freshwater, and terrestrial ecosystems (Basto et al., 2019; Cozzolino et al., 2021; Zeb et al., 2023). Plastic pollution—defined as a contamination that results in or can result in adverse biological effects to resident communities—arises from both the physical impacts, such as

entanglement and ingestion, and the chemical effects, as plastics leach significant amounts of hazardous chemicals (Cozzolino et al., 2023; Nicastro et al., 2023; Thushari and Senevirathna, 2020).

Plastic debris has been classified in many ways over the years, including by size, chemical composition, shape, and colour. Size categories include nano (<1 µm), micro (1 µm –5 mm), meso- (5–25 mm), macro- (25–1000 mm), and mega- (>1 m; Barnes et al., 2009). Microplastics, the most common in marine environments, are further divided into primary (e.g., industrial pellets, microbeads) and secondary types, which result from the breakdown of larger plastics (e.g., Velez et al., 2020). Plastic debris also varies by shape (e.g., fragments, films, foams, pellets, fibres) and colour (Cozzolino et al., 2020; Lusher and Primpke, 2023) with different chemicals often absorbed onto their surfaces (Seuront et al., 2022). Recently, the terminology for plastic debris has expanded to encompass new formations that do not align with previous classifications (Chowdhury et al., 2023; De-la-Torre et al., 2021; Ehlers et al., 2024; Shruti et al., 2023). These formations exhibit altered

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compositions and physical and chemical properties due to interactions with other debris (e.g., brick fragments, glass, sand, wood; [Ellrich et al., 2023](#)) and exposure to anthropogenic and environmental factors (e.g., burning, wave action, weathering; [Rangel-Buitrago and Neal, 2023](#), [Nguyen et al., 2024](#)). Because of their unique characteristics and formation mechanisms, these new types of plastic litter likely demonstrate different transport dynamics, occurrences, and physicochemical properties compared to their original plastic materials, potentially resulting in varied ecological impacts (e.g., [Corcoran and Jazvac, 2020](#)).

One notable new type of plastic debris is pyroplastic, characterised as “an amorphous matrix that appears to be formed by the burning or melting of plastic and that is usually characterised by a single, neutral colour (black-charcoal-grey, off-white or brown), with occasional hues of green, blue, pink or yellow, and is accompanied by cracks and fractures, pits and cavities” ([Turner et al., 2019](#)). Since its initial description six years ago, numerous studies have reported the occurrence of pyroplastics, emphasising the global distribution of this new plastic formation. Pyroplastic have been recorded in the Mediterranean Sea ([Ehlers and Ellrich, 2020](#)), multiple locations in the North and South Atlantic (i.e., Trindade Island, Brazil, Cornwall, UK, and northern France; [Santos et al., 2022](#); [Turner et al., 2019](#); [Zardi et al., 2024](#)), the Bay of Bengal (i.e., Tamil Nadu, India, and Cox's Bazar Beach, Bangladesh; [Gunasekaran et al., 2024](#); [Rakib et al., 2023](#)), and the North and South Pacific (i.e., Lima, Peru, and Honshu, Japan; [De-la-Torre et al., 2022](#); [Furukuma et al., 2022](#)). Despite widespread global attention, no reports of pyroplastic litter have been published from the African continent to date. This highlights the limited research conducted in Africa to monitor and characterise marine plastic debris, which often does not align with global standards ([Jambeck et al., 2018](#); [Naidoo and Rajkaran, 2020](#)).

Here, we provide the first quantitative (number of items and total mass of litter) and qualitative (spectroscopic identification) baseline data on pyroplastics ([Turner et al., 2019](#)), as well as partially burned plastics which, like pyroplastics, have undergone burning or melting but still retain some features of the original manufactured objects. Sampling was conducted opportunistically along >2000 km of South African sandy shores (22 sites; [Fig. 1](#)) during biodiversity surveys of intertidal habitats between April 2022 and September 2023. The search time at each site varied between 16 and 49 min (averaging 21 min), and the distance covered varied between 270 and 1500 m (averaging 623 m). The strength and limitations of opportunistic sampling strategies for assessing plastic litter contamination have been highlighted in numerous studies ([Benjaminsen et al., 2024](#)). South Africa's coastline experiences a wide range of tidal patterns, with tidal ranges varying depending on the location. Predominant wind directions along the coast are south-easterly to south-westerly, with some seasonal variation (e.g., [Jury and Goschen, 2020](#); [Lutjeharms and Anson, 2001](#); [Zardi et al., 2015](#)). These complex dynamics influence the distribution and accumulation of pyroplastics along the coastline. While the non-standardised sampling approach (e.g., time of day, weather, tidal cycles) does not allow for spatiotemporal analysis (e.g., comparisons among sites or seasons), this method provides crucial initial baseline data essential for identifying potential contamination hotspots of pyroplastics and laying the foundation for formulating hypotheses in more structured future research (e.g., [Acampora et al., 2017](#); [Cozzolino et al., 2021](#); [Jossart et al., 2023](#)).

Fifty-seven burned plastic items (18 pyroplastics and 29 partially burned plastics) were collected and subsequently examined in the laboratory (Figs. 1S and 2S). These items exhibited light blue, whitish, and



Fig. 1. Sampling locations described in full in [Table 1](#). Map sourced from Free World Maps.

black hues, reflecting the thermal degradation of plastics, where colour alterations occur due to the breakdown of dyes and polymers under elevated temperatures. While previous studies have focused exclusively on pyroplastics, we recognised the importance of also investigating partially burned plastics. These materials can be considered precursors to fully burned and weathered pyroplastics, and both share a history of burning and melting processes. Such processes alter their physical and chemical structures, which may influence their interactions with the environment and potentially have distinct or cumulative effects on biodiversity. For each item, the maximum length (*x*-axis: the longest axis of the examined piece), maximum height (*y*-axis: the axis perpendicular to the *x*-axis at the thickest part of the examined piece), and maximum width (*z*-axis: the axis perpendicular to the *x*- and *y*-axes at the thickest part of the examined piece) were measured using digital callipers. The mass of each item was assessed to the nearest 0.01 g using an electronic digital balance. Additionally, floating tests were performed in a beaker filled with artificial seawater (33 psu), prepared by dissolving 38.5 g salt in 1 L ultrapure water at room temperature (19 °C). All burned plastic items were less dense than seawater, with masses ranging from 0.23 to 51.5 g (averaging 5.4 g; Table 1).

To accurately differentiate between plastic and natural materials, Fourier-transform infrared (FTIR) microspectroscopy in attenuated total reflectance (ATR) mode was used to identify the polymer types and confirm their classification (e.g., Cozzolino et al., 2021; Seuront et al., 2022). IR-ATR spectra ranging from 500 cm^{-1} to 3500 cm^{-1} were acquired using a Nicolet iN10 Fourier infrared microspectroscopy instrument (Thermal Fisher Scientific Co., USA). A Micro Tip ATR probe was brought into contact with the sample to record the spectra, with air serving as the background spectrum. Recorded spectra were compared against commercial FTIR spectral libraries, including the Hummel Polymer and Additives Library and the FBI fibre library. ATR-FTIR analyses revealed that pyroplastics and partially burned plastics comprised polyethylene (PE, 37 items), polyethylene terephthalate (PET, 12 items), polypropylene (PP, 5 items) and polystyrene (PS, 3 items; Fig. 2S).

Sources of anthropogenic waste, in general, are diverse and include various marine- and land-based activities (Harris et al., 2021; Velez et al., 2020). Complex interactions involving prevailing winds, ocean currents, coastal geography, and demographics play pivotal roles in the dispersal and accumulation of plastic waste over vast distances (Mai et al., 2023; Onink et al., 2021). For pyroplastic pollution, it is difficult to pinpoint the exact source as multiple factors could contribute to its presence. Previous studies have suggested that recreational activities of beachgoers contribute to burned plastics contamination, as evidenced by findings of food packaging, burnt charcoal, and signs of campfires or barbecues (Kuš et al., 2023; Velez et al., 2020). However, our current study did not reveal such trends, aligning with a significant body of literature that challenges the perception of beach users being the main source of plastic litter. Poor inland waste management practices, inadequate disposal infrastructure, and informal settlements, where uncontrolled burning of plastic waste leading to the formation of plastiglomerates, plasticrusts, and pyroplastics. Additionally, maritime vessels might contribute by incinerating waste to reduce onboard volume and manage limited storage capacity. Such variability emphasises the complexity of marine litter dynamics and the challenges in identifying consistent patterns of contamination, even within the same geographic region (Fig. 2).

Contrary to our expectations, we found no apparent correlation between the quantity of burned plastic items and the time or area covered during sampling (Table 1). Overall, polyethylene (PE) was the most widespread polymer across all coastal regions, while other polymers, such as polyethylene terephthalate (PET) and polypropylene (PP), exhibited more scattered and inconsistent distributions. This suggests no clear distinction in the presence of these polymers among different coastal environments. Similarly, we observed no clear correlation between the size or weight of plastic items and their locations, indicating that larger or heavier plastics were not preferentially found in specific

coastal regions, and other factors may influence their distribution (Table 2). Although we anticipated bays to retain more particles due to their sheltered nature (Nicastro et al., 2008), we found no relationship between plastic debris and bay locations. Tidal excursion along the South African coastline varies due to factors such as coastal topography, bathymetry, and local oceanographic conditions. On the west coast, the tidal range exceeds 2 m between Mean Low Water Springs (MLWS) and Mean High Water Springs (MHWS), while locations such as Cape Town and Port Elizabeth have a tidal range of about 1.5 m between MLWS and MHWS. Durban, on the east coast, experiences a larger tidal range of approximately 1.8 m between MLWS and MHWS (Rautenbach et al., 2019). This, again, highlights the absence of any apparent pattern between tidal fluctuations and plastic litter accumulation. Taken together, these findings suggest that other factors, such as local currents, wind patterns, or land-based pollution sources, may play a substantial role in the distribution of plastic debris along the South African coast. The absence of predictable or linear distribution patterns for burned plastic items aligns with global observations, highlighting the patchy and irregular nature of marine litter distribution and emphasising the complexity of factors influencing plastic debris accumulation in coastal environments. We observed a higher incidence of burned plastics, particularly near the Port of Durban, which stands as the largest and busiest shipping terminal in sub-Saharan Africa. This high throughput contributes to a larger amount of plastic waste in the surrounding areas. Elevated densities of burned plastics were also recorded on the beaches of the small coastal towns of Kenton-on-Sea and Port Alfred in the Eastern Cape province of South Africa. These locations are popular tourist destinations, contributing to increased waste generation. Moreover, their proximity (<300 m) to large rivers such as the Bushmans and Kariega Rivers, as well as the Kowie River, suggests that riverine pathways play a crucial role in transporting plastic waste into marine ecosystems (Fig. 3S). These results highlight the diverse sources and pathways through which plastic pollution accumulates in coastal regions, influenced by industrial activities, presence of river outlets, and local tourism. The sources and origins of pyroplastics and partially burned plastics (as for most secondary plastic items) are difficult to determine due to their altered state resulting from burning and/or melting processes, which obscure the original plastic structure. However, microspectroscopy assessments, combined with in situ observations of prevalent types of plastic waste and knowledge of local activities such as tourism, fisheries, shipping, and industry, can provide insights into the sources, origins, and potential impacts of plastic particles (Velez et al., 2019).

Because pyroplastics and partially burned plastics have only recently been reported and described, they are not yet included in established guidelines for monitoring marine litter on beaches (e.g., OSPAR, CCAMLR Marine Debris Program, US National Marine Debris Program). However, their unique formation processes and physical structure needs separate classification within the broader category of plastic debris. The need for a separate categorisation is further supported by recent findings showing that pyroplastic leachates have distinct effects compared to non-burnt plastic on the key adaptive behaviours of coastal ecosystem engineers (Zardi et al., 2024). Specifically, the ability of mussels to move and aggregate is affected by pyroplastic leachates, particularly those from beached pyroplastics as opposed to those that have not been in the marine environment (raw). These effects likely stem from qualitative and quantitative differences in contaminants associated with raw and beached plastics, as well as distinct adsorption-desorption dynamics.

As global plastic contamination continues to rise, baseline data like those presented here are crucial for assessing whether novel plastic debris, such as pyroplastics and partially burned plastics, represent a global concern. While we acknowledge the limitations of our opportunistic sampling, this study provides an important data on these emerging pollutants and underscores the need for future systematic surveys to enhance our understanding and inform mitigation efforts.

Table 1

Description of retrieved items found on sandy beaches. In the first column, numbering corresponds to those represented in the map in Fig. 1. Codes for each item refer to labelled photos in Fig. 1S. The table also includes coordinates, exposure (i.e., bay or open coast), item counts, distance surveyed (metre) and time (minute), dimensions (x, y, z in cm), mass (g), and polymer type. In the last column, **bold** text represents pyroplastics, while non-**bold** text indicates partially burned plastics.

Number. location CODE	Coordinates	Bay/open coast	Item counts	Distance surveyed (m)	Search time (min)	x-Axis length (cm)	y-Axis length (cm)	z-Axis length (cm)	Mass (g)	Polymer type
1. Durban DU	29°50'47.66"S 31°2'15.60"E	Open coast	17	270	25	1:11.7 2:3.9 3:5.7 4:3.1 5:3.2 6:4.3 7: 4.7 8: 5.2 9: 4.6 10:5.7 11:7.1 12:3.8 13:15.9 14:6.7 15:10.3 16:1.9 17:15	1:6.5 2:2.5 3:3 4:2 5:3 6:2.9 7:4.6 8:3.1 9:3 10:4.3 11:3.7 12:3.8 13:8 14:5.2 15:7.1 16:1.7 17: 7.4	1:1.1 2:1.5 3:0.2 4:0.6 5: 1.2 6: 0.8 7: 0.4 8: 0.7 9: 0.8 10: 1 11: 0.4 12: 0.8 13: 1.7 14:2.2 15: 2.1 16:0.5 17:2.9	1: 3.9 2:0.2 3:0.4 4:0.9 5:5.3 6:5.5 7: 1.7 8:1.8 9:1.7 10:8.3 11:1.5 12:2.1 13:7.8 14:6.2 15:14.3 16:0.4 17:37.2	1:PE 2:PE 3:PE 4:PP 5:PE 6:PE 7:PET 8:PE 9:PE 10:PE 11:PE 12:PE 13:PE 14:PE 15:PE 16:PE 17:PET
2. Port St. Johns PJ	31°38'56.38"S 29°31'6.22"E	Open coast	3	1500	15	1:7 2:3.9 3:7	1:3.1 2:2.8 3:3.5	1:0.5 2:0.3 3:0.4	1:2.8 2:0.9 3:8.1	1:PET 2:PE 3:PE
3. Gonubie GO	32°56'5.83"S 28°1'55.96"E	Open coast	0	350	20					
4. East London EL	33°0'35.23"S 27°55'25.26"E	Open coast	2	340	16	1:5.4 2:2.6	1:3 2:1	1:0.8 2:0.7	1:1 2:0.3	1:PS 2:PET
5. Kidds Beach KB	33°8'43.92"S 27°42'5.64"E	Open coast	1	300	15	1:5	1:4.4	1:0.3	1:1.6	1:PET
6. Port Alfred PA	33°35'43.96"S 26°54'43.71"E	Open coast	6	500	15	1:11.4 2:4.9 3:7 4:1.6 5:17.2 6:4.3	1:4.3 2:3 3:4.9 4:1 5:1.3 6:0.4	1:0.6 2:0.3 :30.9 4:0.3 5:0.8 6:0.3	1:16.7 2:10.9 :317.4 4:0.5 5:6.9 6:0.3	1:PE 2:PE 3:PE 4:PET 5:PE 6:PP
7. Kenton on sea KS	33°41'46.33"S 26°39'20.14"E	Open coast	6	600	20	1:3.1 2:3.8 :32.8 4:5.6 5:1.9	1:3 2:2.1 :32.4 4:3.5 5:1.1	1:0.4 2:0.4 :30.4 4:1.4 5:0.3	1:4.4 2:2.4 :31.5 4:10.1 5:0.4	1:PE 2:PET 3:PS 4:PE 5:PP
8. Port Elizabeth PE	33°58'49.87"S 25°39'32.93"E	Bay	0	500	20					
9. Plettenberg Bay PB	34°5'31.12"S 23°22'22.66"E	Bay	2	600	20	1:3.7 2:5.7	1:2.2 2:3.4	1:0.3 2:0.4	1:1 2:2	1:PE 2:PP
10. Wilderness WN	33°59'46.28"S 22°34'33.06"E	Open coast	2	500	30	1:5 2:3.5	1:2.2 2:0.9	1:0.2 2:0.4	1:0.1 2:1	1:PE 2:PE
11. Glentana GL	34°3'3.29"S 22°18'36.44"E	Open coast	3	600	15	1:4.8 2:4 3:3.9	1:3.1 2:3.8 3:2.5	1:0.2 2:0.3 3:0.3	1:0.4 2:1.4 3:0.6	1:PET 2:PE 3:PE
12. Mossel Bay MB	34°9'38.05"S 22°6'39.49"E	Bay	2	400	49	1:3.7 2:2.1	1:3.5 2:1.2	1:0.4 2:0.3	1:2.3 2:0.4	1:PE 2:PE
13. Cape Agulhas CA	34°48'53.80"S 19°56'39.19"E	Open coast	1	600	20	1:4	1:1.5	1:1.4	1:3.7	1:PET
14. Cape Town CT	34°5'57.74"S 18°29'42.49"E	Open coast	0	700	20					
15. Bloubergstrand BB	33°49'19.39"S 18°28'31.27"E	Bay	2	400	17	1:22 2:7	1:7 2:2	1:0.1 2:0.1	1:1.8 2:0.6	1:PE 2:PE
16. Yzerfontein YF	33°21'19.07"S 18°9'10.71"E	Open coast	1	290	20	1:2.7	1:2.3	1:0.4	1:0.9	1:PE
17. Saldanha Bay SB	32°59'46.10"S 17°58'19.82"E	Bay	1	900	25	1:3.9	1:1.3	1:0.1	1:0.2	1:PET
18. Jacobs Bay JB	32°57'46.42"S 17°53'3.13"E	Bay	0	700	15					
19. Elands Bay EB	32°19'0.19"S 18°19'31.94"E	Bay	1	950	20	1:5.5	1:2	EB1:0.1	EB1:0.4	EB1:PE
20. Lamberts Bay LB	32°6'38.14"S 18°18'7.82"E	Bay	0	1000	25					
21. Doring Bay DB	31°49'17.33"S 18°14'4.97"E	Bay	2	1000	25	1:4 2:2.5	1:3.5 2:1	1:0.4 2:0.1	1:9.4 2:0.3	1:PS 2:PP
22. Port Nolloth PN	29°15'16.97"S 16°52'4.35"E	Open coast	6	700	25	1:7.8 2:7 3:4.5 4:3.9 5:3.3 6:5.6	1:5.1 2:3.4 3:3.6 4:2.4 5:2.2 6:4	1:0.9 2:0.5 3:0.5 4:0.4 5:0.4 6:0.9	1:51.5 2:12.2 3:5.5 4:8.6 5:0.7 6:21.3	1:PET 2:PE 3:PE 4:PE 5:PE 6:PET

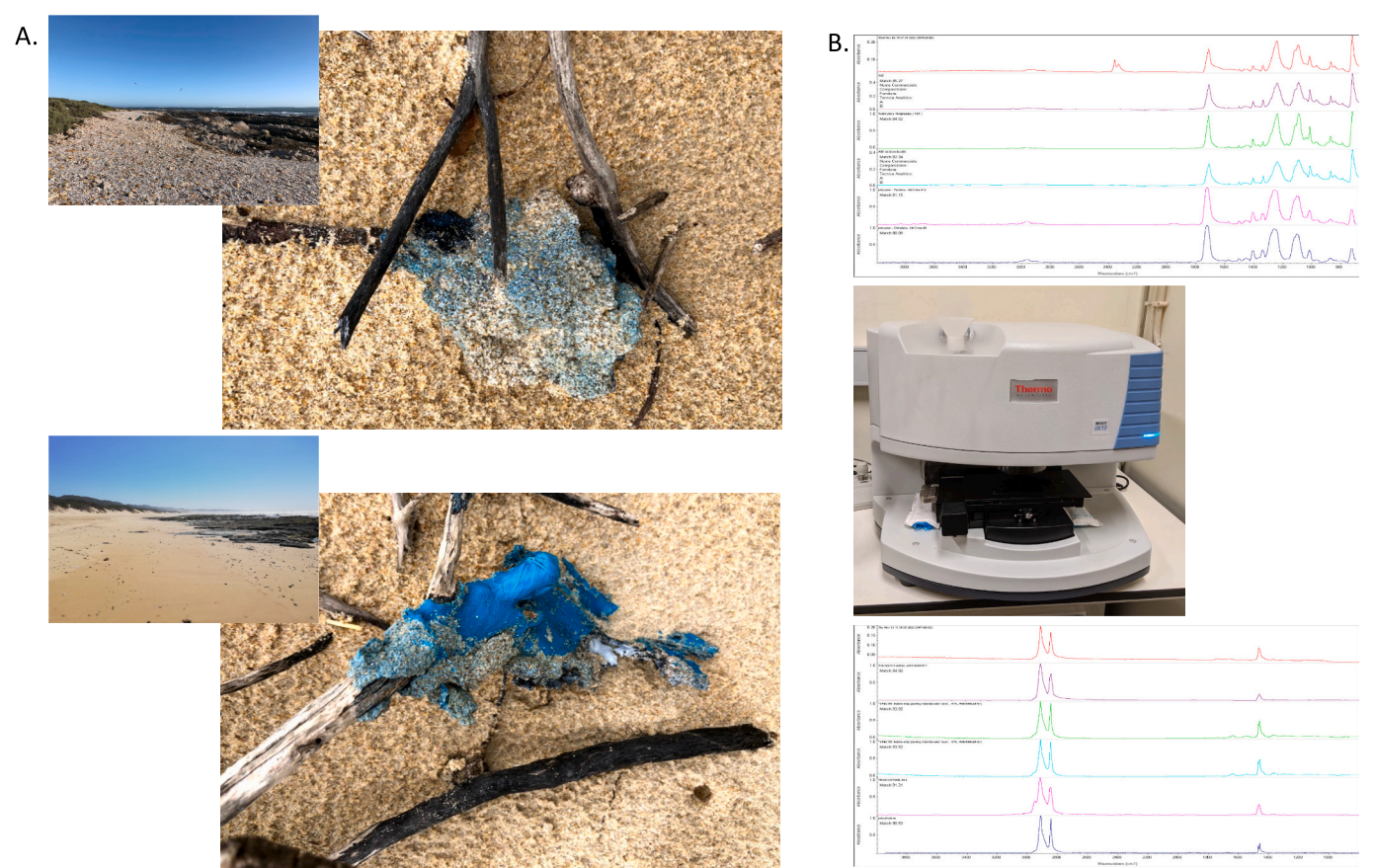


Fig. 2. (A) Pyroplastic items collected at Port Nolloth (top) and Port Alfred (bottom). Insets show the beach environment at each location and a close-up of the pyroplastic fragments. (B) A photo of the Nicolet iN10 Fourier infrared microspectroscopy instrument used for the analysis and examples of FTIR spectra confirming the polymer composition of the pyroplastic samples (photos of additional all items are available in Supplementary material, Fig. S1).

Table 2
Average and standard deviations ($\pm$ SD) of the dimensions for the retrieved items by site. Codes correspond to map 1.

Code	Average x-axis length (cm)	Average y-axis length (cm)	Average z-axis length (cm)	Average mass (g)	SD x-axis length (cm)	SD y-axis length (cm)	SD z-axis length (cm)	SD mass (g)
DU	6.635	4.223	1.111	5.835	4.142	1.955	0.742	8.902
PJ	5.967	3.133	0.400	3.933	1.790	0.351	0.100	3.731
EL	4.000	2.000	0.750	0.650	1.979	1.414	0.071	0.495
KB	5.000	4.400	0.300	1.600	0	0	0	0
PA	7.733	2.480	5.533	58.783	5.675	1.863	12.429	126.850
KS	9.440	8.420	6.580	9.760	13.127	13.436	13.323	12.681
PL	4.700	2.800	0.350	1.500	1.414	0.848	0.071	0.707
WN	4.250	1.550	0.300	0.550	1.061	0.919	0.141	0.636
GL	4.233	3.133	0.267	0.800	0.494	0.650	0.057	0.529
MB	2.900	2.350	0.350	1.350	1.131	1.626	0.071	1.343
CA	4.000	1.500	1.400	3.700	0	0	0	0
BB	14.500	4.500	0.100	1.200	10.606	3.535	0	0.848
YF	2.700	2.300	0.400	0.900	0	0	0	0
SB	3.900	1.300	0.100	0.200	0	0	0	0
EB	5.500	2.000	0.100	0.400	0	0	0	0
DB	3.250	2.250	0.250	4.850	0	0	0	0
PN	5.350	3.450	0.600	16.633	0	0	0	0

CRediT authorship contribution statement

Gerardo I. Zardi: Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Kevin C.K. Ma:** Writing – review & editing, Investigation. **Pierre William Froneman:** Writing – review & editing, Investigation. **Katy R. Nicastro:** Writing – original draft, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by ANR (project SAN22202). GI Zardi is a

laureate of the WINNING Normandy Program, supported by the Normandy Region. This project has received funding from the European Union's Horizon 2020 Research and Innovation Programme under the Marie Skłodowska-Curie grant agreement No. 101034329 and the South African Research Chairs Initiative (SARChI) of the Department of Science and Technology and the National Research Foundation (Grant No. 64801). We also acknowledge the financial support of the Institut de Recherche pour le Développement (IRD), France. We thank Lorenzo Cozzolino and Amy Bernier-Desmarais for their valuable field assistance.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2025.117817>.

Data availability

Data will be made available on request.

References

- Acampora, H., Newton, S., O'Connor, I., 2017. Opportunistic sampling to quantify plastics in the diet of unfledged Black Legged Kittiwakes (*Rissa tridactyla*), Northern Fulmars (*Fulmarus glacialis*) and Great Cormorants (*Phalacrocorax carbo*). *Mar. Pollut. Bull.* 119, 171–174.
- Barnes, D.K., Galgani, F., Thompson, R.C., Barlaz, M., 2009. Accumulation and fragmentation of plastic debris in global environments. *Philos. Trans. R. Soc. Lond. B: Biol. Sci.* 364, 1985–1998.
- Basto, M.N., Nicastro, K.R., Tavares, A.I., McQuaid, C.D., Casero, M., Azevedo, F., Zardi, G.I., 2019. Plastic ingestion in aquatic birds in Portugal. *Mar. Pollut. Bull.* 138, 19–24.
- Benjaminsen, S.C., Dehnard, N., Herzke, D., Johnsen, A., Anker-Nilssen, T., Bourgeon, S., Collard, F., Langset, M., Christensen-Dalsgaard, S., Gabrielsen, G.W., 2024. The challenges of opportunistic sampling when comparing prevalence of plastics in diving seabirds: a multi-species example from Norway. *Mar. Pollut. Bull.* 199, 116037.
- Chowdhury, P.R., Medhi, H., Bhattacharyya, K.G., Hussain, C.M., 2023. Emerging plastic litter variants: a perspective on the latest global developments. *Sci. Total Environ.* 858, 159859.
- Corcoran, P.L., Jazvac, K., 2020. The consequence that is plastiglomerate. *Nat. Rev. Earth Environ.* 1, 6–7.
- Cozzolino, L., Nicastro, K.R., Zardi, G.I., de los Santos, C.B., 2020. Species-specific plastic accumulation in the sediment and canopy of coastal vegetated habitats. *Sci. Total Environ.* 723, 138018.
- Cozzolino, L., de los Santos, C.B., Zardi, G.I., Repetto, L., Nicastro, K.R., 2021. Microplastics in commercial bivalves harvested from intertidal seagrasses and sandbanks in the Ria Formosa lagoon. *Portugal. Mar. Freshw.* 72, 1092–1099.
- Cozzolino, L., Nicastro, K.R., Lefebvre, S., Corona, L., Froneman, P.W., McQuaid, C., Zardi, G.I., 2023. The effect of interspecific and intraspecific diversity on microplastic ingestion in two co-occurring mussel species in South Africa. *Mar. Pollut. Bull.* 196, 115649.
- De-la-Torre, G.E., Dioses-Salinas, D.C., Pizarro-Ortega, C.I., Santillán, L., 2021. New plastic formations in the Anthropocene. *Sci. Total Environ.* 754, 142216.
- De-la-Torre, G.E., Pizarro-Ortega, C.I., Dioses-Salinas, D.C., Rakib, M.R.J., Ramos, W., Pretell, V., Ribeiro, V.V., Castro, Í.B., Dobadarán, S., 2022. First record of plastiglomerates, pyroplastics, and plasticrusters in South America. *Sci. Total Environ.* 833, 155179.
- Ehlers, S.M., Ellrich, J.A., 2020. First record of 'plasticrusters' and 'pyroplastic' from the Mediterranean Sea. *Mar. Pollut. Bull.* 151, 110845.
- Ehlers, S.M., Ellrich, J.A., Gestoso, I., 2024. New Plastic Pollution Types and Novel Sources of Microplastic Pollution in Marine Systems. *Frontiers Media SA*, p. 1372876.
- Ellrich, J.A., Ehlers, S.M., Furukuma, S., Pogoda, B., Koop, J.H.E., 2023. Characterization of three plastic forms: Plasticconcrete, plastimetal and plastisessiles. *Sci. Total Environ.* 895, 165073.
- Europe, P., 2022. The Circular Economy for Plastics – A European Overview.
- Furukuma, S., Ellrich, J.A., Ehlers, S.M., 2022. Frequent observations of novel plastic forms in the Ariho River estuary, Honshu, Japan. *Sci. Total Environ.* 848, 157638.
- Gunasekaran, K., Mghili, B., De-la-Torre, G.E., Sompongchaiyakul, P., Rangel-Buitrago, N., Wang, X., Charoenpong, C., 2024. First record of plastiglomerates, pyroplastics and plasticrusters along the beaches of Tamilnadu, Southeast coast of India. *Mar. Poll. Bull.* 205, 116594.
- Harris, P.T., Westerveld, L., Nyberg, B., Maes, T., Macmillan-Lawler, M., Appelquist, L., 2021. Exposure of coastal environments to river-sourced plastic pollution. *Sci. Total Environ.* 769, 145222.
- Jambeck, J., Hardesty, B.D., Brooks, A.L., Friend, T., Teleki, K., Fabres, J., Beaudoin, Y., Bamba, A., Francis, J., Ribbink, A.J., Baleta, T., Bouwman, H., Knox, J., Wilcox, C., 2018. Challenges and emerging solutions to the land-based plastic waste issue in Africa. *Mar. Policy* 96, 256–263.
- Jossart, Q., Bauman, D., Moreau, C.V.E., Saucède, T., Christiansen, H., Brasier, M.J., Convey, P., Downey, R., Figuerola, B., Martin, P., Norenburg, J., Rosenfeld, S., Verheye, M., Danis, B., 2023. A pioneer morphological and genetic study of the intertidal fauna of the Gerlache Strait (Antarctic Peninsula). *Environ. Monit. Assess.* 195, 514.
- Jury, M.R., Goschen, W.S., 2020. Physical ocean-atmosphere variability over the shelf of South Africa from reanalysis products. *Cont. Shelf Res.* 202, 104135.
- Kuś, S., Jelonek, I., Jelonek, Z., 2023. Effects of thermal treatment of food using barbecue fuels on ambient air and beach sands within recreation facilities. *Sci. Rep.* 13, 17621.
- Lusher, A.L., Primpke, S., 2023. Finding the balance between research and monitoring: when are methods good enough to understand plastic pollution? *Environ. Sci. Technol.* 57, 6033–6039.
- Lutjeharms, J.R.E., Ansgore, I.J., 2001. The Agulhas Return Current. *J. Mar. Syst.* 30 (1–2), 115–138.
- Mai, L., Sun, X., Zeng, E.Y., 2023. Country-specific riverine contributions to marine plastic pollution. *Sci. Total Environ.* 874, 162552.
- Naidoo, T., Rajkaran, A., 2020. Impacts of plastic debris on biota and implications for human health: a South African perspective. *S. Afr. J. Sci.* 116, 1–8.
- Nguyen, M.K., Rakib, M.R.J., Nguyen, H.L., Lin, C., Malafaia, G., Idris, A.M., 2024. A mini-review on plasticrusters: occurrence, current trends, potential threats, and recommendations for coastal sustainability. *Environ. Monit. Assess.* 196, 629.
- Nicastro, K.R., Zardi, G.I., McQuaid, C.D., Teske, P.R., Barker, N.P., 2008. Coastal topography drives genetic structure in marine mussels. *Mar. Ecol. Prog. Ser.* 368, 189–195.
- Nicastro, K.R., Seuront, L., Cozzolino, L., Froneman, W., Zardi, G.I., 2023. The neglected role of intraspecific variation in plastic pollution research. *Anthr. Sci.* 2, 141–147.
- Onink, V., Jongedijk, C.E., Hoffman, M.J., van Seville, E., Laufkötter, C., 2021. Global simulations of marine plastic transport show plastic trapping in coastal zones. *Environ. Res. Lett.* 16, 064053.
- Rakib, M.R.J., De-la-Torre, G.E., Jolly, Y.N., Al Nahian, S., Khan, N.I., Idris, A.M., 2023. First record of plastiglomerate and pyroplastic pollution in the world's longest natural beach. *Sci. Total Environ.* 891, 164369.
- Rangel-Buitrago, N., Neal, W.J., 2023. A geological perspective of plastic pollution. *Sci. Total Environ.* 893, 164867.
- Rautenbach, C., Barnes, M.A., de Vos, M., 2019. Tidal characteristics of South Africa. *Deep-Sea Res. I Oceanogr. Res. Pap.* 150, 103079.
- Santos, F.A., Diório, G.R., Guedes, C.C.F., Bernardino, G., Giannini, P.C.F., Angulo, R.J., de Souza, M.C., César-Oliveira, M.A.F., dos Santos Oliveira, A.R., 2022. Plastic debris forms: Rock analogues emerging from marine pollution. *Mar. Pollut. Bull.* 182, 114031.
- Seuront, L., Zardi, G.I., Uguen, M., Bouchet, V.M.P., Delaeter, C., Henry, S., Spilmont, N., Nicastro, K.R., 2022. A whale of a plastic tale: a plea for interdisciplinary studies to tackle micro- and nanoplastic pollution in the marine realm. *Sci. Total Environ.* 846, 157187.
- Shruti, V., Kutralam-Muniasamy, G., Pérez-Guevara, F., 2023. New forms of particulate plastics in the anthropocene. *Earth Sci. Rev.* 104601.
- Thushari, G.G.N., Senevirathna, J.D.M., 2020. Plastic pollution in the marine environment. *Heliyon* 6, e04709.
- Tong, X., Jong, M.-C., Zhang, J., You, L., Gin, K.Y.-H., 2021. Modelling the spatial and seasonal distribution, fate and transport of floating plastics in tropical coastal waters. *J. Hazard. Mater.* 414, 125502.
- Turner, A., Wallerstein, C., Arnold, R., Webb, D., 2019. Marine pollution from pyroplastics. *Sci. Total Environ.* 694, 133610.
- Velez, N., Zardi, G.I., Lo Savio, R., McQuaid, C.D., Valbusa, U., Sabour, B., Nicastro, K.R., 2019. A baseline assessment of beach macrolitter and microplastics along northeastern Atlantic shores. *Mar. Pollut. Bull.* 149, 110649.
- Velez, N., Nicastro, K.R., McQuaid, C.D., Zardi, G.I., 2020. Small scale habitat effects on anthropogenic litter material and sources in a coastal lagoon system. *Mar. Pollut. Bull.* 160, 111689.
- Windsor, F.M., Durance, I., Horton, A.A., Thompson, R.C., Tyler, C.R., Ormerod, S.J., 2019. A catchment-scale perspective of plastic pollution. *Glob. Chang. Biol.* 25, 1207–1221.
- Zardi, G.I., Nicastro, K.R., McQuaid, C.D., et al., 2015. Intraspecific genetic lineages of a marine mussel show behavioural divergence and spatial segregation over a tropical/subtropical biogeographic transition. *BMC Evol. Biol.* 15, 100.
- Zardi, G.I., Seuront, L., Spilmont, N., Froneman, P.W., Nicastro, K.R., 2024. Leachates from pyroplastics alter the behaviour of a key ecosystem engineer. *Estuar. Coast. Shelf Sci.* 301, 108740.
- Zeb, A., Liu, W., Ali, N., Shi, R., Wang, Q., Wang, J., Li, J., Yin, C., Liu, J., Yu, M., 2023. Microplastic pollution in terrestrial ecosystems: global implications and sustainable solutions. *J. Hazard. Mater.* 132636.